SparseRank: Explainable Cost-Aware Feature Selection for Anomaly Detection in E-Commerce Microservices*

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Abstract

Modern e-commerce platforms run on microservice architectures that emits massive stream of system-level signals. Collecting, storing, and analysing all of them is expensive and crucial to identify the root cause in case of abnormal behaviour. We present SparseRank, a lightweight framework that presents which monitoring signals actually matter for anomaly detection in a microservice of E-Commerce business. SparseRank reduces monitored feature space by 94% while maintaining detection quality by combining sparse feature selection with an unsupervised anomaly detector and a feature-attribution layer tested on a real benchmark that covers fourteen fault situations across twelve services.

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